

LAB ASSIGNMENT – 4

Course: Fundamentals of Operating System Lab (ENCA252)

Program: BCA (AI & DS)

Title: Implementation and Analysis of Disk Scheduling Algorithms

Aim

To implement and analyze disk scheduling algorithms including FCFS, SSTF, SCAN, and C-SCAN, and compare their performance based on total seek time.

Objective

- To understand disk scheduling concepts
- To simulate disk head movement
- To calculate total seek time
- To compare efficiency of algorithms

Tools Used

- Python 3.x
- Ubuntu/Linux
- VS Code

Theory

Disk scheduling algorithms determine the order in which disk I/O requests are serviced. Efficient scheduling reduces seek time and improves overall system performance.

Algorithms Implemented

1. FCFS (First Come First Serve)

- Requests are processed in arrival order
- Simple but inefficient

2. SSTF (Shortest Seek Time First)

- Selects the request closest to current head
- Reduces seek time

3. SCAN (Elevator Algorithm)

- Head moves in one direction servicing requests
- Reverses direction at end

4. C-SCAN (Circular SCAN)

- Head moves in one direction only
- Jumps back to start

Input

- Disk Request Queue:

98, 183, 37, 122, 14, 124, 65, 67

- Initial Head Position: 53

- Disk Size: 200

Output

Algorithm	Seek Time
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FCFS	640
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SSTF	236
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SCAN	331
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C-SCAN	382
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Analysis

- SSTF gives minimum seek time because it selects nearest request
- FCFS is simplest but gives maximum seek time
- SCAN provides balanced performance

- C-SCAN gives uniform wait time but slightly higher seek time

Result

The program successfully implemented all disk scheduling algorithms and calculated total seek time for each.

Conclusion

Disk scheduling significantly affects system performance. SSTF is efficient for reducing seek time, while SCAN and C-SCAN provide fairness and better overall performance.

```
priyanshu7027@clouddevops:~$ python3 disk_scheduling.py
Requests: [98, 183, 37, 122, 14, 124, 65, 67]
Initial Head: 53

FCFS Seek Time: 640
Sequence: [53, 98, 183, 37, 122, 14, 124, 65, 67]

SSTF Seek Time: 236
Sequence: [53, 65, 67, 37, 14, 98, 122, 124, 183]

SCAN Seek Time: 331
Sequence: [53, 65, 67, 98, 122, 124, 183, 199, 37, 14]

C-SCAN Seek Time: 183
Sequence: [53, 65, 67, 98, 122, 124, 183, 199, 0, 14, 37]
priyanshu7027@clouddevops:~$
```
